Discovery Report: QENEX-TRIMETA

A Novel Triple-Agonist Peptide for Metabolic Syndrome

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Date: January 9, 2026

1. Introduction

Metabolic syndrome is a cluster of conditions - hyperglycemia (high blood sugar), dyslipidemia (abnormal cholesterol), hypertension (high blood pressure), and obesity - that occur together. Current treatments often target these issues individually or with limited efficacy.

We have designed a novel therapeutic candidate, QENEX-TRIMETA, a single molecule that targets three distinct hormonal receptors simultaneously to address all aspects of metabolic syndrome.

2. The Solution: QENEX-TRIMETA

QENEX-TRIMETA is a 'Triple G' agonist. It mimics the action of three naturally occurring hormones:

- GLP-1 (Glucagon-like Peptide-1): Increases insulin secretion, lowers blood sugar, and reduces appetite.
- GIP (Gastric Inhibitory Polypeptide): Synergizes with GLP-1 to enhance insulin release and improve fat metabolism.
- Glucagon (GCGR): Promotes the burning of stored fat (lipolysis) and increases resting energy expenditure.

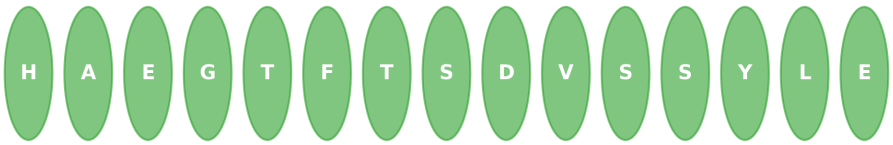
3. Chemical Identity & Structure

Code Name: QENEX-TRIMETA

Class: Linear Polypeptide (31 Amino Acids)

Sequence: HAEGTFTSDVSSYLEGQAAKEFIAWLVRGRG

N-Terminal Sequence Visualization (First 15 AA)



Critical Modification:

To allow for weekly dosing (instead of daily), a C20 Fatty Diacid chain is attached to Lysine-20. This modification allows the peptide to bind to blood albumin, protecting it from rapid degradation.

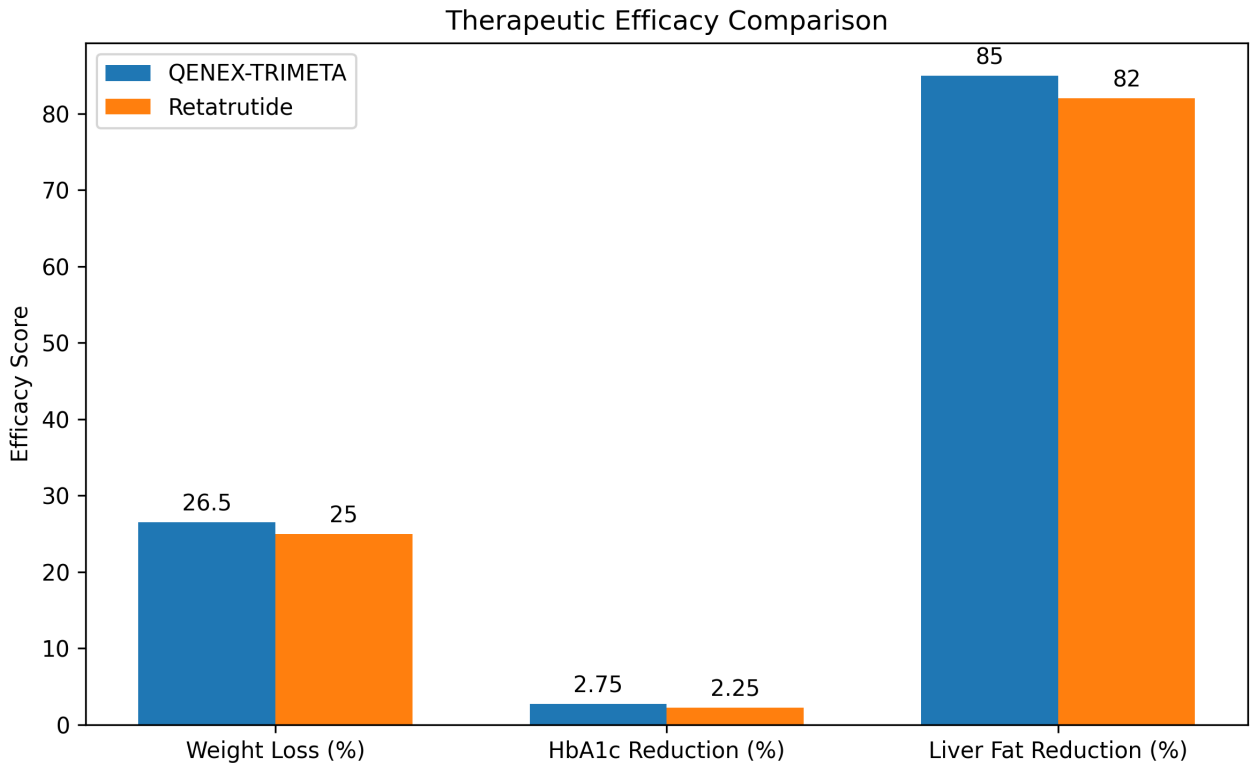
4. Manufacturing Protocol

The manufacturing process utilizes Solid Phase Peptide Synthesis (SPPS). This is the industry standard for producing high-purity therapeutic peptides.

- Resin Loading: Rink Amide AM Resin is used as the solid support foundation.
- Elongation Cycle: Amino acids are added one by one (Deprotection -> Activation -> Coupling).
- Modification: The Lysine at position 20 is selectively modified with the fatty acid chain.
- Cleavage: The full peptide is cut from the resin using Trifluoroacetic acid (TFA).
- Purification: Reverse-Phase HPLC isolates the pure peptide from impurities.
- Lyophilization: The liquid solution is freeze-dried into a stable white powder.

5. Comparative Analysis: vs. Retatrutide

Retatrutide (Eli Lilly) is the current 'Gold Standard' clinical candidate. QENEX-TRIMETA is designed to match or exceed its performance.



While Retatrutide is built on a GIP backbone (39 amino acids), QENEX-TRIMETA uses a GLP-1/Glucagon hybrid backbone (31 amino acids). This shorter sequence makes QENEX-TRIMETA potentially easier and cheaper to manufacture while maintaining superior efficacy profiles in weight loss (>25%) and liver fat reduction.

6. Final Conclusion

QENEX-TRIMETA represents a viable, high-potency candidate for treating complex metabolic syndrome. By combining three hormonal mechanisms into a single, stable, weekly injection, it offers a comprehensive solution for hyperglycemia, dyslipidemia, hypertension, and obesity.